Exercises: Risk-Neutral Pricing and the Vasicek Model

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Six progressive exercises lead from a one-period guarantee to the Vasicek bond price and its use in the thesis. Each builds on the previous one. Together they test the distinction between the real-world measure $\mathbb{P}$ (projection) and the risk-neutral measure $\mathbb{Q}$ (valuation).

Work through them in order. Hints are given; full solutions are withheld unless requested.

Exercise 1: Discrete warm-up (the guarantee as a put)

A one-period market has a risk-free rate of 2% and a risky asset with $S_0 = 100$. At $t = 1$ the asset is worth 110 (up) or 95 (down). Under the real-world measure, $\mathbb{P}(\text{up}) = 0.7$.

1. Find the probability $q = \mathbb{Q}(\text{up})$ under which the discounted asset price S_t/B_t is a martingale.
2. An employer guarantees that the account will be worth at least $G = 100$ at $t = 1$. The employer's top-up is $\max(G - S_1, 0)$. Compute its price as a discounted $\mathbb{Q}$-expectation.
3. Compute the discounted $\mathbb{P}$-expectation of the same payoff. Which number is the cost at which the guarantee could be hedged, and which is the expected cost? Why do they differ?
4. Compute the Radon–Nikodym derivative $Z = d\mathbb{Q}/d\mathbb{P}$ in both states. Verify that the $\mathbb{P}$-expectation of Z equals 1, and interpret which state $\mathbb{Q}$ overweights.

Hint for 1: write the no-arbitrage condition below and solve for q.

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S_0=\frac{\mathbb E^{\mathbb Q}[S_1]}{1.02}
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Exercise 2: Which processes are $\mathbb{Q}$-martingales?

Work in the Vasicek model with bank account $B(t) = \exp(\int_0^t r_s\, ds)$. For each process, decide whether it is a $\mathbb{Q}$-martingale and justify the answer in one or two sentences.

1. $B(t)$
2. $P(t, T)$

3. P(t, T) / B(t)
4. r_t
5. $W_t^{\mathbb{Q}}$
6. The conditional $\mathbb{Q}$-expectation of X / B(T) given $\mathcal{F}_t$, for an integrable payoff X paid at T

Hint: a process with a predictable drift is not a martingale. For item 6, use the tower property.

Exercise 3: Solving the Vasicek SDE

1. Apply Itô's formula to $f(t, r) = e^{\kappa t} r$ and show that the solution is:

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r_T=r_t\,e^{-\kappa(T-t)}+\theta\big(1-e^{-\kappa(T-t)}\big)+\sigma\int_t^T
e^{-\kappa(T-s)}\,\mathrm dW_s
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2. Using the properties of the Itô integral, give the conditional mean and variance of r_T given $\mathcal{F}_t$ under $\mathbb{P}$.
3. What changes under $\mathbb{Q}$? What stays the same?
4. What happens to the mean and variance as $T \to \infty$? Relate the answer to footnote 1 of the blog post.

Hint for 2: the integrand $e^{-\kappa(T-s)}$ is deterministic, so the integral is normally distributed. Its variance follows from the Itô isometry.

Exercise 4: Deriving the bond price

Write $\tau = T - t$ and use the function:

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B(\tau)=\frac{1-e^{-\kappa\tau}}{\kappa}
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1. Integrate the solution of Exercise 3 (under $\mathbb{Q}$) over [t, T] and show that, given $\mathcal{F}_t$:

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\mathbb E^{\mathbb Q}\Big[\int_t^T r_s\,\mathrm ds\,\Big|\,\mathcal
F_t\Big]=\theta^{\mathbb Q}\tau+(r_t-\theta^{\mathbb Q})\,B(\tau)
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2. Argue why $\int_t^T r_s \, ds$ is conditionally normally distributed. Its conditional variance is given below. Deriving it is optional but good practice with the Itô isometry.

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V(\tau)=\frac{\sigma^2}{\kappa^2}\Big(\tau-B(\tau)-\frac{\kappa}
{2}B(\tau)^2\Big)
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3. For Y normally distributed with mean m and variance v, the expectation of e^{-Y} equals $e^{-m + v/2}$. Use this to obtain $P(t, T) = A(\tau)\, e^{-B(\tau)\, r_t}$. Check the resulting $\ln A(\tau)$ against the formula in the blog post.

Hint for 1: the stochastic integral has mean zero, so taking expectations before integrating over time avoids the double integral.

Exercise 5: Sanity checks

1. Verify that $P(T, T) = 1$.
2. Show that the yield $y(t, T)$ converges, as $\tau \to \infty$, to the long yield below.
3. The long yield lies *below* $\theta^{\mathbb{Q}}$. Explain intuitively why higher volatility lowers long yields.

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y_\infty=\theta^{\mathbb Q}-\frac{\sigma^2}{2\kappa^2}
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Hint for 3: $x \mapsto e^{-x}$ is convex. Apply Jensen's inequality to the discount factor.

Exercise 6: Interpretation for the thesis

1. In a 20-year projection used to train the agent, which long-run level, θ or $\theta^{\mathbb{Q}}$, governs the simulated path of r_t? Which one enters the yield curve computed at each simulated date? Explain why.
2. Suppose $\lambda < 0$, so investors demand a premium for interest-rate risk. Is $\theta^{\mathbb{Q}}$ above or below θ? What does this imply for simulated yields compared with simulated short rates?
3. Valuing a path-dependent guarantee at future dates generally requires nested simulation: outer paths under $\mathbb{P}$, inner valuations under $\mathbb{Q}$. Explain why this is unnecessary for bond prices in the Vasicek model. For which quantities in the plan-design problem would it still be needed?
4. If the 2.5% statutory guarantee were valued with $\mathbb{P}$-expectations instead of $\mathbb{Q}$-expectations, would the employer's guarantee cost typically be over- or understated? Use Exercise 1 to argue.